

Relational Model

Remember??

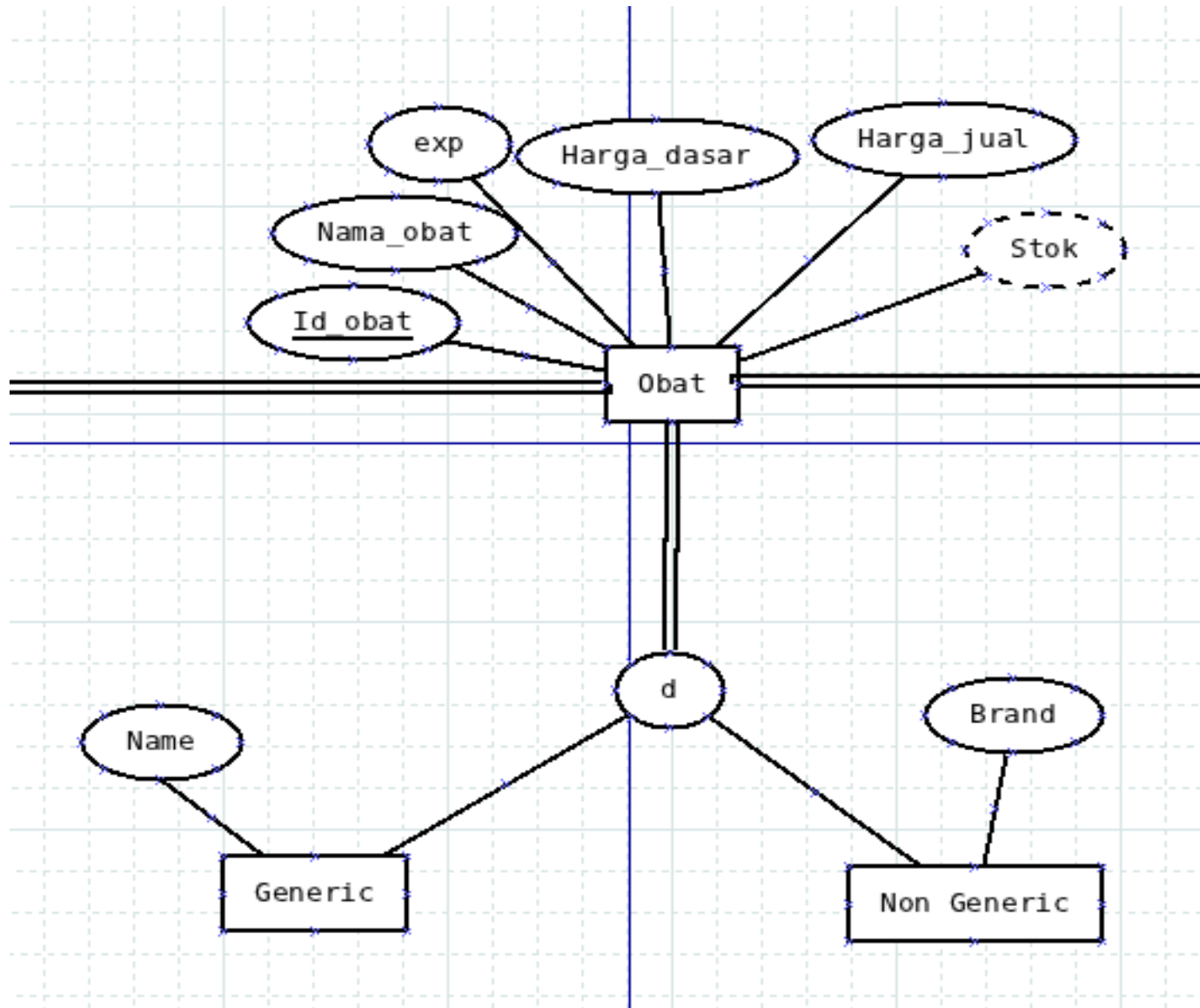
A. ER to Realtional Model

1. Mapping of Regular Entity Types
2. Mapping of Weak Entity Types
3. Mapping of Binary 1:1 Relationship Types
4. Mapping of Binary 1:N Relationship Types.
5. Mapping of Binary M:N Relationship Types.
6. Mapping of Multivalued attributes.
7. Mapping of N-ary Relationship Types.

B. E-ER to Relational Model

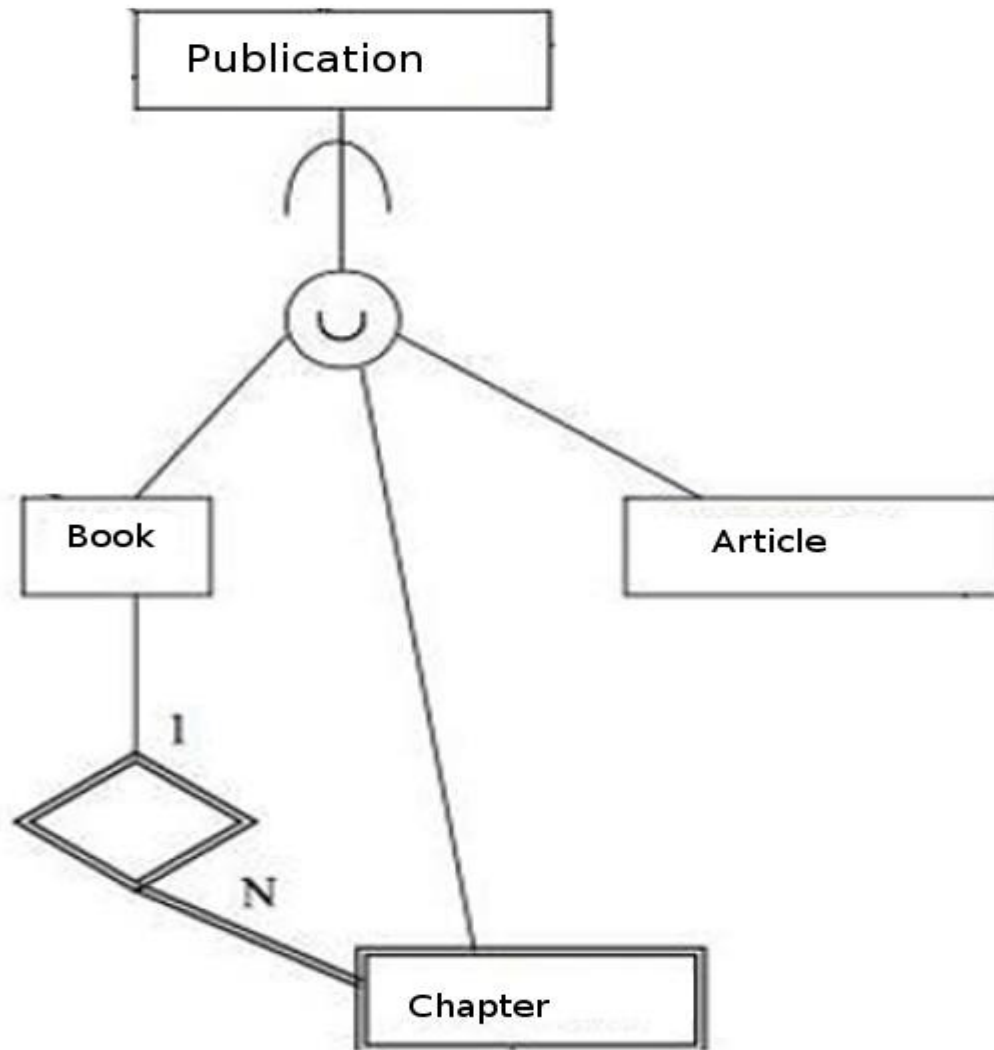
8. Options for Mapping Specialization or Generalization.
9. Mapping of Union Types (Categories).

Exercise



**Create Relational
Model
With 4 options**

Exercise



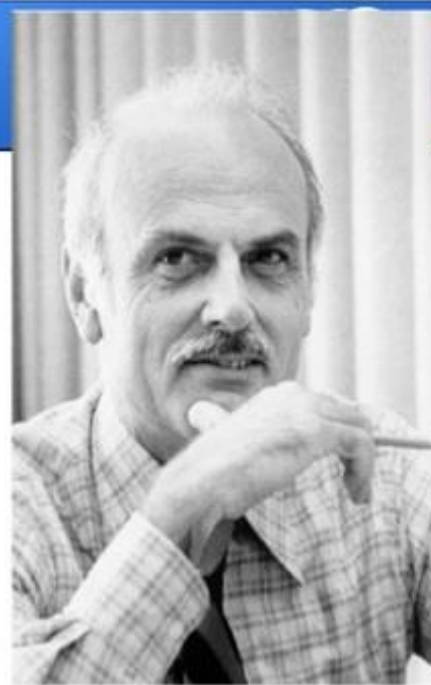
Relational Model Concepts

The model was first proposed by Dr. E.F. Codd of IBM in 1970 in the following paper:

"A Relational Model for Large Shared Data Banks," Communications of the ACM, June 1970

**Edgar Frank
"Ted" Codd**

**Inventor of
Relational Model for
Database**



Relational Model Concepts

- The relational Model of Data is based on the concept of a Relation.
- A Relation is a mathematical concept based on the ideas of sets.
- **The strength** of the relational approach to data management comes from the formal foundation provided by the theory of relations.

Relations

- Relational DBMS products store data in the **form of relations**, a special type of **table**
- A **relation** is a **two-dimensional table** that has the following characteristics
 - **Rows** contain data **about an entity**
 - **Columns** contain data about **attributes of the entity**
 - **Cells** of the table hold a **single value**
 - **All entries** in a column are of **the same kind**
 - Each **column** has a **unique** name
 - **The order** of the **columns** is **unimportant** – concept of set (x)
 - **The order** of the **rows** is **unimportant** (x)
 - **No two rows** may be identical (x)

Relations

- **Set of Tuples** and Typically Shown as a **Table** With **Columns** and **Rows**.
- **Column (Field)** Represents an **Attribute**
- **Row (Tuple)** Represents an **Entity Instance**

relation name

R

Tuples

t_1

t_2

t_m

A1	A2		An
v_{11}	v_{12}		v_{1n}
v_{21}	v_{22}		v_{2n}
$\vdots$			
v_{m1}	v_{m2}		v_{mn}

$t_1[A_n]$

Formal Definitions

- A **Relation** may be defined in multiple ways.
- The **Schema** of a Relation: $R (A1, A2, \dots, A_n)$
Relation schema R is defined over **attributes** $A1, A2, \dots, A_n$

For Example -

CUSTOMER (Cust-id, Cust-name, Address, Phone#)

Here, **CUSTOMER** is a relation defined over the **four attributes** Cust-id, Cust-name, Address, Phone#, each of which has a **domain or a set of valid values**. For example, the domain of Cust-id is 6 digit numbers.

Formal Definitions (2)

- A **domain** has a logical definition: e.g., “USA_phone_numbers” are the set of 10 digit phone numbers valid in the U.S.
- A **domain** may have a **data-type** or a **format defined** for it. The USA_phone_numbers may have a format: (ddd)-ddd-dddd where each d is a decimal digit. E.g., Dates have various formats such as monthname, date, year or yyyy-mm-dd, or dd mm,yyyy etc.
- An **attribute** designates the **role** played by the domain. E.g., the domain Date may be used to define attributes “Invoice-date” and “Payment-date”.

Formal Definitions (3)

- Notation:

- We refer to **component values** of a tuple t by $t[A_i] = v_i$ (the value of attribute A_i for tuple t).

Similarly, $t[A_u, A_v, \dots, A_w]$ refers to the subtuple of t containing the values of attributes $A_u, A_v, \dots, A_w$, respectively.

Formal Definitions (3)

- Let $S1 = \{0,1\}$
- Let $S2 = \{a,b,c\}$
- Let $R \subseteq S1 \times S2$
- Then for example: $r(R) = \{ \langle 0,a \rangle , \langle 0,b \rangle , \langle 1,c \rangle \}$
is one possible “state” or “population” or “extension” r of the relation R , defined over domains $S1$ and $S2$. It has three tuples.

Relations

Figure 4.2 Sample Relation

EmployeeNumber	FirstName	LastName	Department	Email	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	236-9987
200	Mary	Abernathy	Finance	MA@somewhere.com	444-8898
300	Liz	Smathers	Finance	LS@somewhere.com	777-0098
400	Tom	Caruthers	Accounting	TC@somewhere.com	236-9987
500	Tom	Jackson	Production	TJ@somewhere.com	444-9980
600	Eleanore	Caldera	Legal	EC@somewhere.com	767-0900
700	Richard	Bandalone	Legal	RB@somewhere.com	767-0900

Not Relations

Figure 4.3b Tables but Not Relations — Multiple Entries per Cell

EmployeeNumber	FirstName	LastName	Department	Email	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	236-9987
200	Mary	Abernathy	Finance	MA@somewhere.com	444-8898
300	Liz	Smathers	Finance	LS@somewhere.com	777-0098
400	Tom	Caruthers	Accounting	TC@somewhere.com	236-9987
				Fax:	266-9987
				Home:	555-7171
500	Tom	Jackson	Production	TJ@somewhere.com	444-9980
600	Eleanore	Caldera	Legal	EC@somewhere.com	767-0900
				Fax:	236-9987
				Home:	555-7171
700	Richard	Bandalone	Legal	RB@somewhere.com	767-0900

(b)

Relation Keys

- Example
 - EMP(ENO, ENAME, TITLE, SAL)
 - PROJ (PNO, PNAME, BUDGET)
 - WORKS(ENO, PNO, RESP, DUR)
- **Underlined Attributes are Relation Keys** which Uniquely Distinguish Among Tuples (Rows)
- Tabular Form

EMP	<u>ENO</u>	ENAME	TITLE	SAL
PROJ	<u>PNO</u>	PNAME	BUDGET	
WORKS	<u>ENO</u>	<u>PNO</u>	RESP	DUR

Relation Instances

EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

WORKS

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

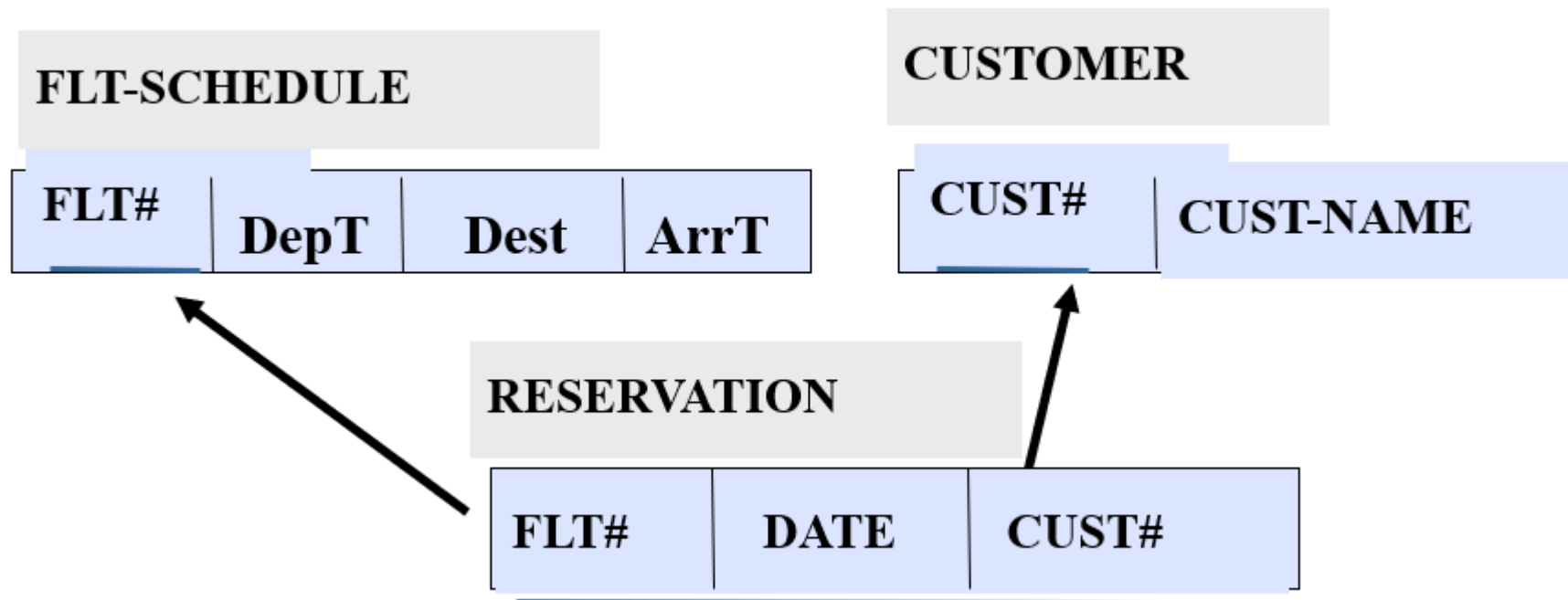
PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Characteristics of Attribute

- ❑ An Attribute Name Refers to a **Position in a Tuple by Name** Rather than Position
- ❑ An Attribute Name Indicates the **Role of a Domain** in a Relation
- ❑ Attribute Names must be **Unique Within Relations**
- ❑ By Using Attribute Names we **can Disregard** the Ordering of Field Values in Tuples
- ❑ Can Be a **Null** Value Meaning “Not Known”, “Not Applicable” ...

Relational Integrity Constraints

- **IC:** Conditions that Must **Hold on All Valid Relation Instances** at Any Given Database State. Why are Integrity Constraints Needed? What Happens when we try to Delete a Flight?



Relational Integrity Constraints

- There are three **main types** of constraints:
 - **Key** constraints
 - **Entity integrity** constraints
 - **Referential integrity** constraints
- Other Types of **Semantic** Constraints:
 - **Domain** Constraints
 - **Transition** Constraints
 - **Set** Constraints
- DBMSs Handle Some But Not All Constraints

Key Constraints

- **Superkey (SK):**
 - Any Subset of Attributes Whose Values are Guaranteed to **Distinguish Among Tuples.**
 - Sample:
 {PlayerId}, {Email}, {PhoneNumber}, {PlayerId, Email}
 {PlayerId, PhoneNumber}, {PlayerId, Email, DOB}

PlayerId	FirstName	LastName	Email	PhoneNumber	DOB (Date of Birth)
1	Nathan	Tjoe	Nathan.Tjoe@gmail.com	555-1234	1/1/1990
2	Maarten	Paes	Maarten.Paes@gmail.com	555-5678	5/15/1991
3	Jay	Idzes	Jay.Idzes@gmail.com	555-9876	8/20/1992
4	Mees	Hilgers	Mees.Hilgers@gmail.com	555-9879	1/1/1990
5	Maarten	Hilgers	Maarten.Hilgers@gmail.c	555-9873	8/20/1992

Key Constraints

- **Candidate Key (CK):**
 - A **Superkey** with a **Minimal Set of Attributes** (No Attribute Can Be Removed Without Destroying the Uniqueness -- Minimal Identity)
 - A **Value of an Attribute** or a Set of Attributes in a Relation That Uniquely Identifies a Tuple
 - There may be **Multiple Candidate Keys**
 - It cannot have unnecessary
 - Sample:
 {PlayerId}, {Email}, {PhoneNumber}

PlayerId	FirstName	LastName	Email	PhoneNumber	DOB (Date of Birth)
1	Nathan	Tjoe	Nathan.Tjoe@gmail.com	555-1234	1/1/1990
2	Maarten	Paes	Maarten.Paes@gmail.com	555-5678	5/15/1991
3	Jay	Idzes	Jay.Idzes@gmail.com	555-9876	8/20/1992
4	Mees	Hilgers	Mees.Hilgers@gmail.com	555-9879	1/1/1990
5	Maarten	Hilgers	Maarten.Hilgers@gmail.com	555-9873	8/20/1992

Key Constraints

- **Primary Key (PK):**

- Choose **One** From Candidate Keys
- The Primary Key Attributed are **Underlined**
- Sample:

{PlayerId} or {Email} or {PhoneNumber} “**Choose One**”

PlayerId	FirstName	LastName	Email	PhoneNumber	DOB (Date of Birth)
1	Nathan	Tjoe	Nathan.Tjoe@gmail.com	555-1234	1/1/1990
2	Maarten	Paes	Maarten.Paes@gmail.com	555-5678	5/15/1991
3	Jay	Idzes	Jay.Idzes@gmail.com	555-9876	8/20/1992
4	Mees	Hilgers	Mees.Hilgers@gmail.com	555-9879	1/1/1990
5	Maarten	Hilgers	Maarten.Hilgers@gmail.c	555-9873	8/20/1992

Key Constraints

● Composite Key (PK):

- Composite Key is a key made up of more than one attribute to uniquely identify a row.
- Usually found in tables whose keys are a combination of FK
- Sample:
{ENO, PNO}

WORKS			
ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

Key Constraints

● Foreign Key (FK):

- An Attribute or a **Combination of Attributes** (Say A) of Relation R1 Which Occurs **as the Primary Key of another** Relation R2 (Defined on the Same Domain)
- **Allows Linkages Between Relations** that are Tracked and Establish **Dependencies**
- Useful to Capture ER Relationships

Schema With Key

CAR	<u>LicenseNumber</u>	EngineSerialNumber	Make	Model	Year
	Texas ABC-739	A69352	Ford	Mustang	96
	Florida TVP-347	B43696	Oldsmobile	Cutlass	99
	New York MPO-22	X83554	Oldsmobile	Delta	95
	California 432-TFY	C43742	Mercedes	190-D	93
	California RSK-629	Y82935	Toyota	Camry	98
	Texas RSK-629	U028365	Jaguar	XJS	98

Super Key?

Candidate Key?

Composite Key?

Primary Key?

Determine Key type

EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

WORKS

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
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E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Super Key?

Candidate Key?

Composite Key?

Primary Key?

The diagram illustrates the following relationships:

- EMPLOYEE** (FNAME, MINIT, LNAME, SSN, BDATE, ADDRESS, SEX, SALARY, SUPERSSN, DNO)
 - Has a self-referencing relationship on SSN and SUPERSSN.
 - Is linked to **DEPARTMENT** via DNAME and DNUMBER.
 - Is linked to **DEPT_LOCATIONS** via DNUMBER and DLOCATION.
 - Is linked to **PROJECT** via PNUMBER and DNUM.
 - Is linked to **WORKS_ON** via ESSN and PNO.
 - Is linked to **DEPENDENT** via ESSN and DEPENDENT_NAME.
- DEPARTMENT** (DNAME, DNUMBER, MGRSSN, MGRSTARTDATE)
 - Is linked to **EMPLOYEE** via DNAME and DNUMBER.
 - Is linked to **DEPT_LOCATIONS** via DNUMBER and DLOCATION.
 - Is linked to **PROJECT** via PNUMBER and DNUM.
- DEPT_LOCATIONS** (DNUMBER, DLOCATION)
 - Is linked to **DEPARTMENT** via DNUMBER and DLOCATION.
 - Is linked to **PROJECT** via PNUMBER and DNUM.
- PROJECT** (PNAME, PNUMBER, PLOCATION, DNUM)
 - Is linked to **EMPLOYEE** via PNUMBER and DNUM.
 - Is linked to **WORKS_ON** via ESSN and PNO.
- WORKS_ON** (ESSN, PNO, HOURS)
 - Is linked to **EMPLOYEE** via ESSN and PNO.
- DEPENDENT** (ESSN, DEPENDENT_NAME, SEX, BDATE, RELATIONSHIP)
 - Is linked to **EMPLOYEE** via ESSN and DEPENDENT_NAME.

Coresponding DB Table

EMPLOYEE	FNAME	MINIT	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John		Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin		Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia		Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer		Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh		Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce		English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad		Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James		Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

					DEPT_LOCATIONS	<u>DNUMBER</u>	<u>DLOCATION</u>
DEPARTMENT	DNAME	<u>DNUMBER</u>	MGRSSN	MGRSTARTDATE			Houston
							Stafford
							Bellaire
							Sugarland
	Research	5	333445555	1988-05-22			
	Administration	4	987654321	1995-01-01			
	Headquarters	1	888665555	1981-06-19			

You can find location and startdate by employee's SSN value

Corresponding DB Table

WORKS_ON	<u>ESSN</u>	<u>PNO</u>	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	<u>PNUMBER</u>	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

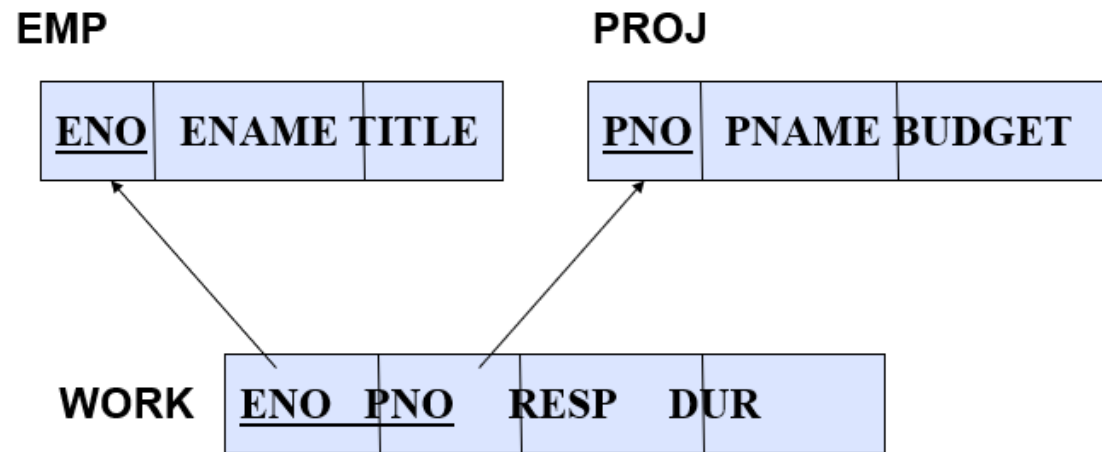
DEPENDENT	<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Entity Integrity

- **Relational Database Schema:**
 - A **Set S** of Relation Schemas $(R_1, R_2, \dots, R_n)$ That Belong to the **Same Database**
 - **S** is the **Name of the Database**
 - **$S = \{R_1, R_2, \dots, R_n\}$**
- Entity Integrity:
 - For **Any R_i in S** , **Pk_i is the Primary Key of R**
 - Attributes in **Pk_i Cannot Have Null Values** in any Tuple of $R(r_i)$
 - **$T[Pk_i] \neq \text{Null}$ for Any Tuple T in $R(r)$**

Referential Integrity

- A Constraint Involving *Two Relations* Used **to Specify a Relationship** Among Tuples in
- Referencing Relation and Referenced Relation
- A Referential Integrity Constraint **Can Be Displayed in a Relational Database Schema** as a **Directed Arc** From R_1 .**FK** to R_2 .**PK**



WORK[ENO] is a subset of EMP[ENO]

WORK[PNO] is a subset of PROJ[PNO]

Operations on Relations

- A **DBMS Operates via User Queries** to Read and Change Data in a Database
- Changes Can be Inserting, Deleting, or Updating (Equivalent to a Delete followed by Insert)
- One Critical Issue in DB Operations is Integrity Constraints Maintenance in the Presence of
 - INSERTING a Tuple
 - DELETING a Tuple
 - UPDATING/MODIFYING a Tuple.

Operations on Relations

- Integrity constraints should not be violated by the update operations.
- Several update operations may have to be grouped together.
- Updates may *propagate* to cause other updates automatically. This may be necessary to maintain integrity constraints.

Update Operations on Relations

- Integrity constraints should not be violated by the update operations.
- Several update operations may have to be grouped together.
- Updates may *propagate* to cause other updates automatically. This may be necessary to maintain integrity constraints.

Inserting Operations

- Insert a **Duplicate Key Violates Key Integrity**:
 - Check If Duplicates Occur
- Insert a **Null Key Violates Entity Integrity**:
 - Check If Null is in Any Key
- Insert a **Tuple Whose Foreign Key Attribute Pointing to an Non-existent Tuple Violates Referential Integrity**:
 - Check the Existence of Referred Tuple

Deleting Operations (sample)

EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

WORKS

ENO	PNO	RESP	DUR
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E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Deletion Operations

- Deleting a **Tuple Referred to by Other Tuples in Database (via FKs) would Violate Referential Integrity**
- Action:
 - Check for Incoming Pointers of the Deleted Tuple.
 - Group the Deletion and the Post-processing of the Referencing Pointers in a Single Transaction

Deletion Operations

- Three Options If Deletion Causes a Violation
 - Reject the Deletion
 - Attempt to Cascade (Propagate) the Deletion by Deleting the Tuples which Reference the Tuple being or to be Deleted
 - Modify the Referencing Attribute Values that Cause the Violation; Each Values is Set to Null or Changed to Reference to Another Valid Tuple

Inserting Operations (sample)

EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
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PNO	PNAME	BUDGET
P1	Instrumentation	150000
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P3	CAD/CAM	250000
P4	Maintenance	310000
P5	CAD/CAM	500000

Modify Operations

- Modify Operation **Changes Values of One or More Attributes in a Tuple (or Tuples)** of a Given Relation R
- Maintaining ICs Requires to Check If the Modifying Attributes Are Primary Key or Foreign Keys.

Modify Operations

Integrity Check Actions:

- Case 1:

If the Attributes to be Modified are **Neither a Primary Key nor a Foreign Key**, Modify Causes **No Problems**

Must Check and Confirm that the New Value is of **Correct Data Type and Domain**

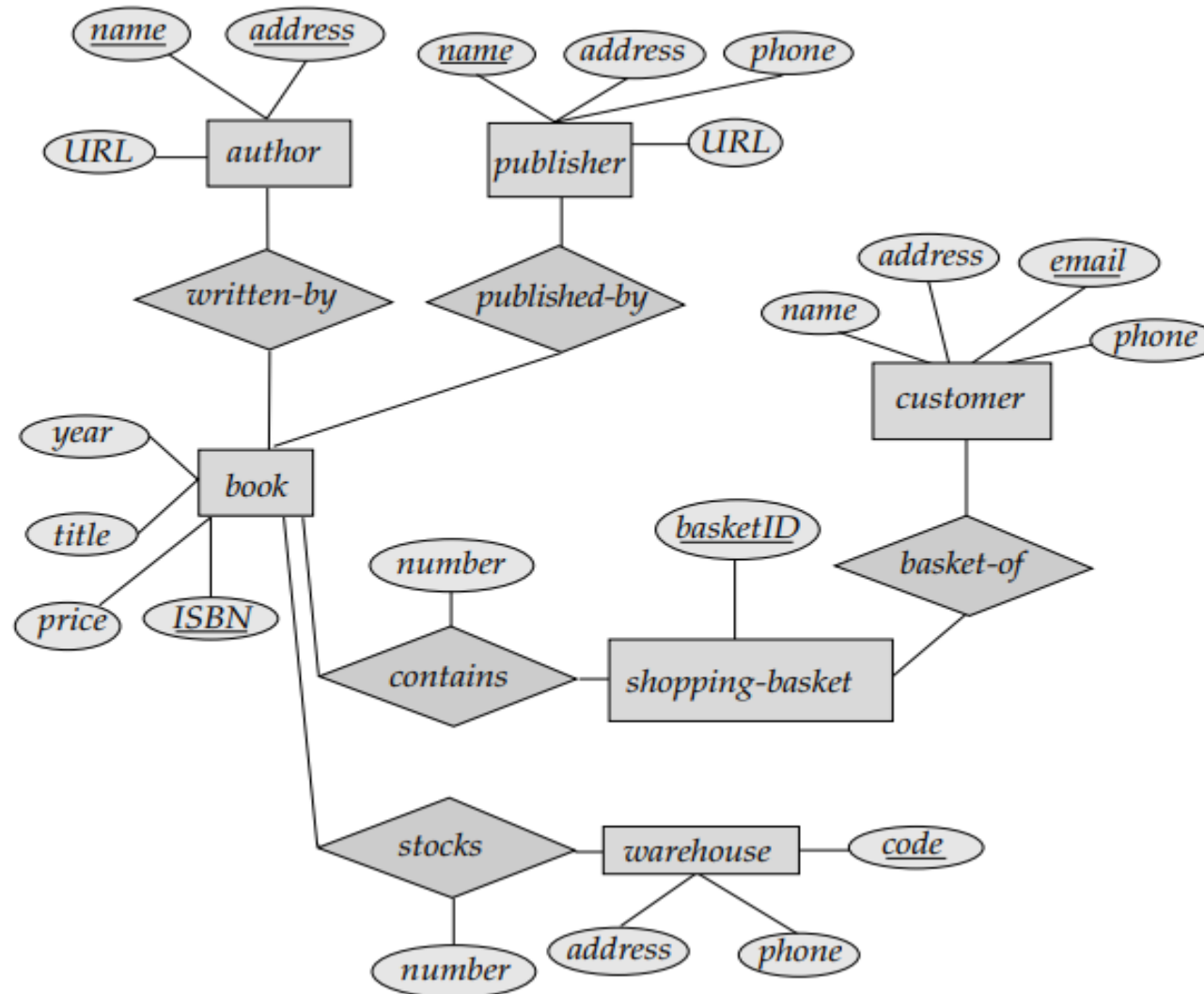
- Case 2:

Modifying a Primary Key Value Similar to Deleting One Tuple and Insert Another in its Place

Summary

- 1. Key**
- 2. Operation Insert**
- 3. Operation Delete**
- 4. Operation Update**

Exercise





Barangsiapa yang berangkat menimba ilmu untuk mengamalkan ilmu,
niscaya ilmu yang sedikit pun akan bermanfaat baginya